

Practice exercise, Lecture 4

ØKA2015 Applied Data Science. Not graded. Do it before Tuesday.

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Reference: the Lecture 4 slides, `lecture04_wrangling.R`, and *R for Data Science* Chapter 3 up to “Rows”. Use the `flights` table from the `nycflights13` package, loaded with `library(tidyverse)` and `library(nycflights13)`.

Work in a script. For each task, write the command, run it with **Ctrl + Enter**, and read the first line of the output: the row count tells you whether the condition did what you meant. Nine tasks, about 50 minutes.

Task 1: operators and the pipe

- Predict, then run.

```
5 > 2 & 3 == 4
"EWR" %in% c("JFK", "LGA")
!(10 < 20)
```

- Rewrite each line with the pipe, and say in words what it does.

```
nrow(flights)
filter(flights, dest == "BOS")
```

Task 2: filter, one condition

- Flights in December.
- Flights that arrived more than one hour late.
- Flights with a missing arrival delay. What do they have in common?

Task 3: filter, several conditions

- Flights to Boston ("BOS") or Washington ("DCA"). Two spellings; write both.
- Flights in June, July or August that left JFK.
- Flights that arrived more than two hours late but did not leave late.
- Flights delayed by at least an hour that made up more than 30 minutes in flight.

Task 4: the two mistakes

- Run `flights |> filter(month = 12)`. What does R say, and what did it want?
- Run `flights |> filter(month == 12 | 1)` and count the rows. Why that number? Write the version that does what was meant.

Task 5: arrange

- The ten flights with the longest arrival delay. Which airline, which day?
- The flights that left earliest in the morning. What does `dep_time` 1 mean?
- Flights sorted by distance, shortest first. Where does the shortest go?
- The ten longest flights by `air_time`. Where do they go?

Task 6: distinct and count

- How many different airlines are there? How many different destinations?
- How many flights did each airline operate, largest first?
- Was there a flight on every day of 2013? Use `distinct()` and `nrow()`.
- Which origin airport had the most flights?
- Does it matter whether you `filter()` first and then `arrange()`, or the other way round? Try both on “December flights, longest departure delay first”, and think about which order does less work.

Task 7: a pipeline you can read aloud

Write one pipeline, one verb per line, that takes the flights, then keeps December, then keeps arrivals more than an hour late, then puts the worst first. Read it aloud as a sentence. Then store it as `december_late` and check its row count.

Task 8: another dataset

Download the *Global Superstore* dataset from Kaggle (search for it; it is free). It has columns such as Order ID, Order Date, Customer Name, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit. Read it with `read_csv()`, then:

- Find all orders where the category is "Technology" and the discount is greater than 0.2.
- Find the five orders with the highest profit.
- How many unique customers are in the dataset? How many orders did each region have, largest first?

Task 9: save your work

Put every command you used into a script, with a comment above each one saying what it does, and save it as `YourName_lecture04.R`. Close RStudio, open it again, and run the whole script from the top. It should work without you typing anything else. If it does not, the missing line is one of the two `library()` calls.

Answers to “Check yourself” from the script

```
flights |> filter(arr_delay >= 120)           # 1: 10,200 rows
flights |> filter(dest %in% c("IAH", "HOU"))   # 2: 9,313
flights |> filter(carrier %in% c("UA", "AA", "DL")) # 3: 139,504
flights |> filter(month %in% c(7, 8, 9))       # 4: 86,326
flights |> filter(arr_delay > 120 & dep_delay <= 0) # 5: 29
flights |> arrange(desc(dep_delay))           # 6: 1,301 minutes, 9 January
flights |> arrange(dep_time)                  #   dep_time 1 is 00:01
flights |> distinct(year, month, day) |> nrow() # 7: 365, so yes
flights |> arrange(desc(distance)) |> glimpse() # 8: JFK to Honolulu, 4,983 miles
flights |> arrange(distance) |> glimpse()      #   Newark to LaGuardia, 17 miles
```

The printout shows only the columns that fit, so `dest` and `distance` are often hidden. `|> glimpse()` shows every column of the top rows; `View()` opens the whole table.

The answers to Tasks 1 to 9 are not distributed. Compare with a classmate, or bring your script to Tuesday's lecture.